Lab 44: Retry when the format is wrong, stop when the data is absent

Topic 4: Prompt Engineering & Structured Output · Task 4.4: Implement validation, retry, and feedback loops for extraction quality

Runs on Simulator 6: Structured Data Extraction · Run in class

Source: Built from exam guide task statement 4.4 (no practice question covers this angle)

The question

Validation fails on two documents: a date written DD/MM, and a receipt whose total doesn't match its line items. Which is worth a retry?

A. Resend the document, bad output and error. Stop if the source lacks it.

B. Retry everything until something comes back.

Simulate it live

```
# In the browser: open Simulator 6, then Lab 44
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 44

# In a terminal, from simulators/simulator-6-structured-extraction/
uv run extract.py
```

What you should see

DOC-03 keeps both totals and goes to a human. DOC-07 names what is missing and goes to a human without a retry. A format error gets a retry.

Open `simulators/simulator-6-structured-extraction/evidence/` and find it yourself.

The answer

A. Resend the document, bad output and error. Stop if the source lacks it.

Now change one thing

Change DOC-04's printed total in fixtures/invoices.txt, rerun, and watch by_document_type.invoice fall while the others hold.